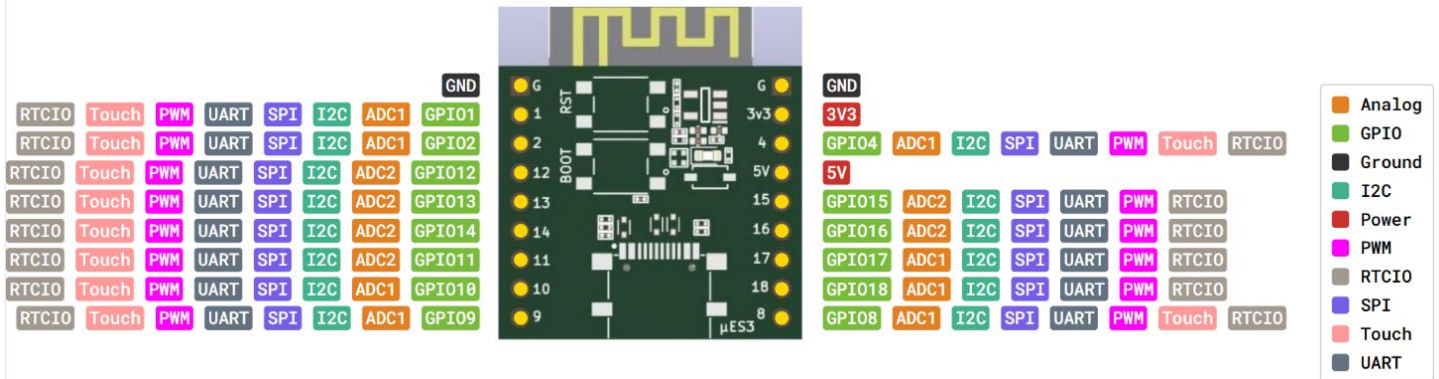


μES3 ESP32-S3 WROOM1 N16R8 WiFi/Bluetooth Microcontroller Dev Board

- **Arduino IDE:** Choose “ESP32S3 Dev Module” in “Tools – Board – esp32”.
- Pinout and Interfacing details are given in below image. 5V pin is power input with a Schottky diode to shield from USB-C VBUS pin. 3.3V pin is the LDO output.



Datasheet: documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

Specifications:

Board dimension	2.7 x 2.4 cm
Processor	Xtensa Dual core 32-bit LX7 microprocessor with max frequency 240 MHz
ROM/SRAM	384 KB ROM/512 KB SRAM on-chip
Flash Memory/PSRAM	16 MB/8 MB
Wireless Communication	2.4 GHz Wi-Fi (802.11b/g/n), Bluetooth 5.0 (BLE)
IO Pins	14 GPIO pins, 5V power pin and 3.3V LDO output pin exposed.
Interface support	I2C, SPI, 12-bit ADC – 14 Inputs, Touch, PWM, LCD, UART, USB 2.0 On-the-Go (OTG) support
Built-in LEDs	GPIO5 for control/status LED. Separate LED for power indication
Antenna	On-board PCB antenna
USB Programming	USB-C port with native USB programming
Switches	On-board BOOT and RESET buttons

Product Overview: μES3 is a high-performance, tiny microcontroller development board based on



ESP32-S3 WROOM 1 N16R8 module. The board runs at 240 MHz, has *16 MB flash memory, 8 MB PSRAM and supports 2.4 GHz WiFi, Bluetooth 5 (BLE) communication*. Programming is done via on-chip/native USB-C without using an external USB-UART bridge chip.

Applications: μES3 is an embedded edge category high performance device that is highly suited for

Speech/Voice recognition, Face recognition, Smart IOT, Wearable applications, Human-machine interface (HMI), Offline AI/ML, Neural network acceleration etc.



MII INITIATIVE (Mu-ETronics Design)